

Technology News

Caretaker robot

Researchers at RIKEN and Tokai Rubber Industries (TRI) have developed a robot that assists in lifting the elderly. Called RIBA-II, the robot uses high-precision tactile sensors and flexible motor-control technology. Developed with the goal of providing high-quality care to the elderly, it can lift up to 80 kg weighing people off floor-level bedding into a wheelchair.

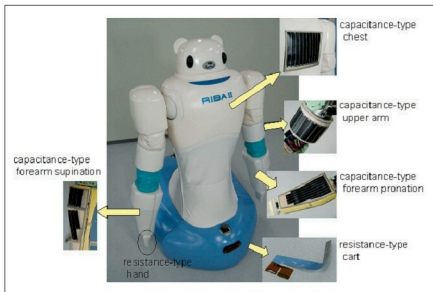
The joints in the robot's base and lower back enable RIBA-II to crouch down and lift a person off a futon at floor level. It accomplishes this task using smart rubber sensors—the first capacitance-type tactile sensors made entirely of rubber. Printed in sheets and fitted onto the robot's arms and chest, the sensors enable high-precision tactile guidance and allow RIBA-II to quickly detect a person's weight from touch alone.

Micro-electronic stimulators for paralysed muscles

Researchers at New Jersey Institute of Technology (NJIT) have developed a wireless device called FLAMES to help victims of spinal cord injury.

The tiny semiconductor device will eventually enable people with spinal cord injuries to restore some of their motor functions lost due to an injury. Energised by an infrared light beam through an optical fibre located just outside the spinal cord, these micro-stimulators activate the nerves in the spinal cord below the point of injury and thus allow the use of the muscles that were once paralysed.

FLAMES is remotely controlled by an optical fibre attached to a low-power near-infrared laser. The device is implanted into the spinal cord, and then allowed to float in the tissue. There are no wires attached. A patient pushes a button on the external unit to activate the laser, which then activates the device.



Locations of smart rubber sensors in RIBA-II caretaker robot that can lift elderly off the floor

"The implanted stimulators are very small, in the sub-millimetre range. A key benefit is that since our device is wireless, the connections can't deteriorate over time, plus the implant causes minimal reaction in the tissue, which is a common problem with similar wired devices," said Mesut Sahin, PhD, associate professor, department of biomedical engineering, NJIT.

Earthquake detector

Researchers at the Indian Institute of Technology, Gandhinagar (IIT-GN) have teamed up with California Institute of Technology (CIT), Pasadena, US, to develop a 'slight early' earthquake warning system.

This system will have a dense network of low-cost motion sensors capable of sensing earthquakes' early seismic activity. Small devices called accelerometers will be put in place near an active fault line, which can pick up earth vibrations. These can be standalone devices transmitting data to a central server.

The sensors' mesh in that area shall be able to pick up the velocity of shock

waves, issuing slight early alerts of an earthquake.

Smartphone to do your diagnostic tests

A team of researchers led by Ki Chon, professor and head of biomedical engineering at Worcester Polytechnic Institute (WPI), has developed a smartphone application that can measure not only the user's heart rate but also heart rhythm, respiration rate and blood oxygen saturation using the phone's built-in video camera.

The iPhone-inspired application yields vital signs as accurate as with standard medical monitors used clinically. The application analyses video clips recorded as the patient's fingertip is pressed against the lens of the phone's camera. As the camera's light penetrates the skin, it reflects off pulsing blood in the finger. The application is able to correlate subtle shifts in the colour of the reflected light with changes in the patient's vital signs. The team created and adapted algorithms to process the data captured by the phone's video camera.